

Demand-Sensing Volatility Score

Four-expert review, revised specification, and empirical validation against 2023–2025 customer demand

DFUs analysed: **12,000** | calendar months: **36** | scorable: **9,105** | routed to manual review (small sample): **2,895**

Executive summary

The proposed design is well grounded — the $ADI=1.32$ and $CV^2=0.49$ cutoffs are the Syntetos–Boylan–Croston demand-classification boundaries, and the repo's *demand_classifier.py* already uses them. The issues below are about how the four metrics are **combined**, not whether they are the right metrics.

- **Redundancy (highest impact).** ADI and ZeroShare are the same variable: $ADI = 1/(1-ZeroShare)$. On the real data their Spearman correlation is **1.000**. Equal 25% weights therefore give intermittency ~50% and quantity-variability ~50% — not four independent quarters. The revised score keeps ADI and drops ZeroShare from the weighted sum (still reported).
- **Spike fragility.** max/mean is sensitive to one outlier and grows with history length. Replaced with robust p95/median.
- **CV^2 saturation.** the $0.25 \rightarrow 0.49$ band is only 0.24 wide and pegs at 1 almost immediately; widened to $0.25 \rightarrow 1.0$ so it discriminates across the erratic range.
- **Seasonality.** predictable seasonal SKUs (zero in the off-season) score high on ADI+ZeroShare and get wrongly flagged high-risk. Added a seasonal-strength guard that exempts/deseasonalises them.
- **Small samples.** with <3 non-zero months CV^2 /spike are unstable; those DFUs are routed to manual review rather than given a false score.

The four-expert review

1 — *Intermittent-demand statistician*

- $ADI \equiv 1/(1 - \text{ZeroShare})$: same signal, two scales. Confirmed on data (Spearman 1.000).
- CV^2 and SpikeRatio also overlap (both measure non-zero dispersion; Spearman 0.88). Four 'independent' metrics are really ~2 factors.
- Fix: one intermittency term (ADI), keep CV^2 + a de-correlated spike term; state the real weighting.

2 — *Forecasting practitioner*

- Seasonality is read as volatility; the most forecastable seasonal SKUs get pushed to manual review.
- MTD-vs-LY must be like-for-like: this year's elapsed days vs the SAME elapsed days last year (MTD-to-MTD), not partial month vs full prior-year month, or early-month signals read low and over-trigger.

3 — *Normalization & robustness reviewer*

- Make every subscore an explicit clamped linear ramp.
- Widen CV^2 top to 1.0; replace max/mean spike with p95/median.
- Guard: <3 non-zero months → not scorable. Fix typos: header 'ASI'→ADI; Spike ' $\geq 2 = 2$ ' → ' $= 1$ '.

4 — *Calibration & productionization*

- 0.33/0.66 are placeholders — re-derive the low-risk band as the score range where automated adjustment historically matched/beat baseline (backtest by score decile).
- Separate 'volatility' from 'confidence' (data sufficiency + signal size) as two gates.
- Log all four subscores per DFU in UAT, not just the total.

Revised specification

Subscore convention unchanged: 0 = stable, 1 = highly volatile; final score in [0,1].

```
GUARDS
  n_nonzero < 3          -> not scorable, route to manual_review
  strong seasonality     -> deseasonalise / exempt (band = low)

RAW METRICS (on trailing monthly series incl. zero months)
  ADI          = n_periods / n_nonzero
  ZeroShare    = n_zero / n_periods          # reported only, NOT scored
  CV2          = (std_nz / mean_nz)^2        # ddof=1, non-zero months
  SpikeRatio   = p95(nonzero) / median(nonzero)

SUBSCORES (clamped linear ramp 0..1)
  intermittency_sub = clamp((ADI - 1.32) / (6.0 - 1.32), 0, 1)
  cv2_sub          = clamp((CV2 - 0.25) / (1.0 - 0.25), 0, 1)
  spike_sub        = clamp((SpikeRatio - 1.0) / (2.0 - 1.0), 0, 1)

SCORE (ZeroShare dropped; weights rebalanced)
  score = 0.4*intermittency_sub + 0.35*cv2_sub + 0.25*spike_sub

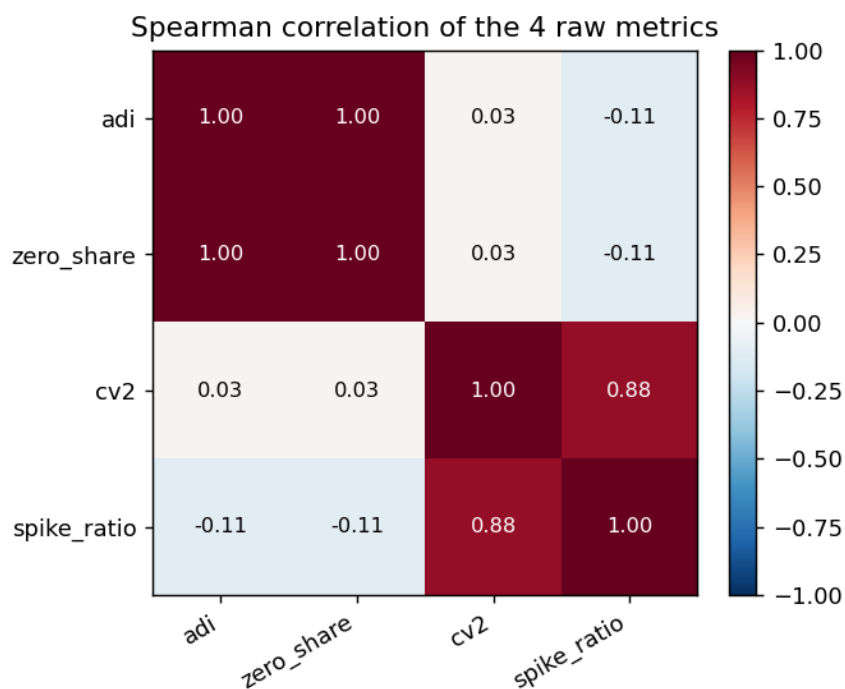
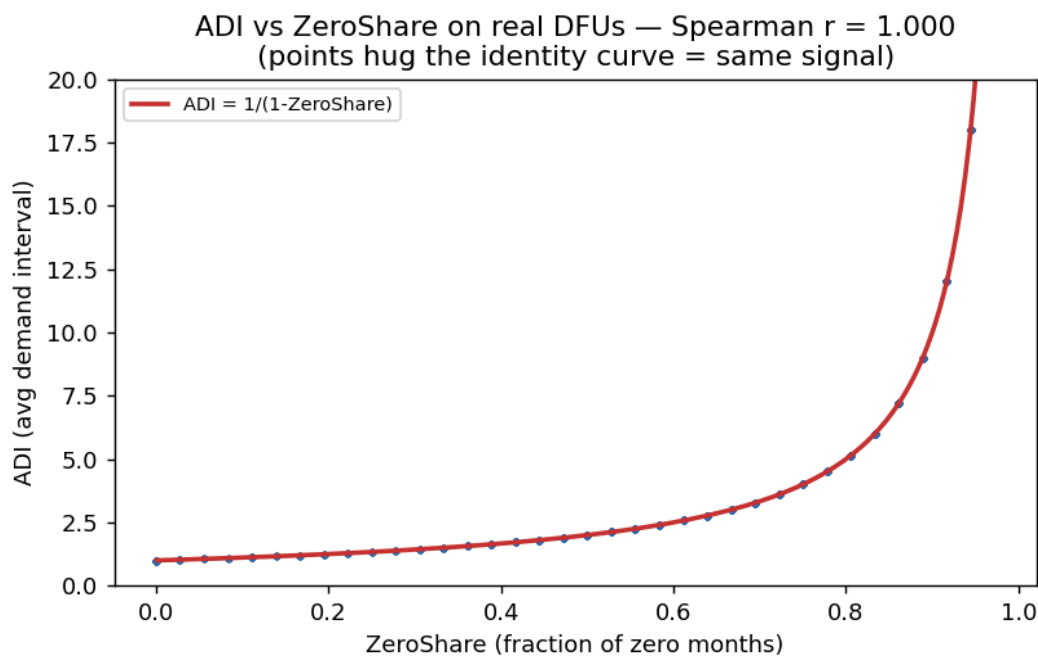
BANDS (UAT default – calibrate against backtested error)
  0.00-0.33 low  (automate)   0.34-0.66 medium  0.67-1.0 high
```

MTD-vs-LY decision logic (like-for-like):

```
ratio = MTD_consumption_this_year / MTD_consumption_same_elapsed_days_last_year
if band == 'low':      auto-apply forecast adjustment from ratio
elif band == 'medium': adjust but flag for review / damp the change
else (high/manual):   no auto-change; planner review
```

Empirical validation (real 2023–2025 demand)

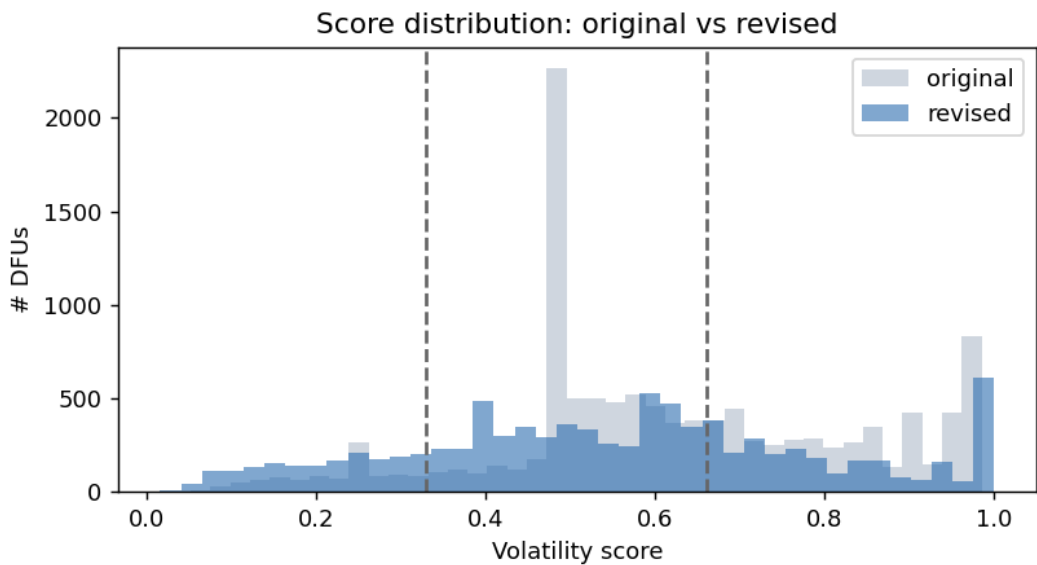
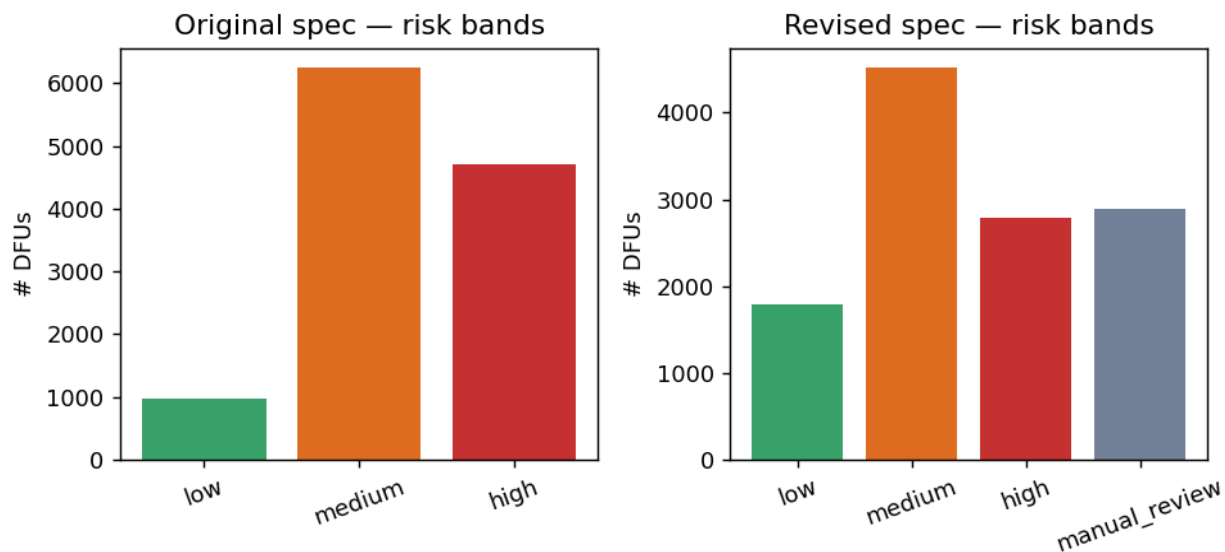
Finding 1 — ADI and ZeroShare are redundant. Spearman correlation 1.000; points sit on the algebraic curve $ADI = 1/(1 - ZeroShare)$. Including both double-counts intermittency.



CV² and SpikeRatio also correlate, confirming the four metrics collapse to ~two underlying factors.

Impact on risk bands & automation rate

Re-weighting and the seasonality / small-sample guards shift how many DFUs are eligible for automation (the 'low' band) and isolate the truly hard ones.



Band	Original	Revised
low	973	1788
medium	6254	4522
high	4711	2795
manual_review	—	2895

Recommendations & open questions

1. Drop ZeroShare from the weighted sum (keep ADI); report ZeroShare for transparency.
2. Use p95/median for spike intensity; widen the CV² ramp to 0.25→1.0.
3. Compare MTD-to-MTD on equal elapsed days, not partial-vs-full-month.
4. Add the seasonality guard so predictable seasonal SKUs stay automatable.
5. Route <3-non-zero-month DFUs to manual review (low confidence ≠ low volatility).
6. Backtest-calibrate the 0.33/0.66 cut points against realized forecast error per decile.
7. Treat 'confidence' (history length + signal size) as a second gate alongside volatility.

Open questions for the team

- Lookback window for the score — exactly 12 months, or trailing 24–36 for stability?
- DFU grain: is site+warehouse+item correct, or should customer be included/excluded?
- Should medium-risk auto-adjust with damping, or always go to a planner?
- Final weights: keep 0.40/0.35/0.25, or fit them to minimize backtested error?

Generated from `analysis_volatility/build_report.py` · `volatility_score.py` is the reusable scoring module.